

POINT HOPE, AK, USA

WMO: 701043

Lat: 68.350N

Lon: 166.800W

Elev: 13

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26623

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-21.7	-29.2	1.1	-14.9	-28.5	1.1	-16.9	46.9	-0.4	42.7	1.8	9.8	320	1.089

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	58.8	53.6	55.1	51.2	53.7	50.2	54.4	57.9	52.0	55.0	50.2	52.9	10.4	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.4	58.4	56.7	49.8	53.1	53.9	47.8	49.2	52.2	22.9	57.9	21.3	55.1	20.2	52.8	60.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
40.1	35.4	32.0	-31.8	64.1	6.5	5.0	-36.5	67.7	-40.3	70.6	-43.9	73.4	-48.7	77.0		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	-1.2	-0.9	1.5	11.7	29.0	38.4	46.3	46.4	40.8	28.3	16.3	4.1	
	DBStd	20.27	13.24	13.30	12.95	11.16	7.23	5.89	5.23	4.18	5.78	8.24	9.58	11.58	
	HDD50	10321	1587	1425	1503	1150	651	354	140	126	278	673	1011	1423	
	HDD65	15748	2052	1845	1968	1600	1116	799	580	576	725	1138	1461	1888	
	CDD50	48	0	0	0	0	0	5	24	16	3	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	15.3	15.7	15.5	15.0	13.6	12.2	12.4	13.5	15.1	16.0	17.6	18.3	18.5	
Precipitation	PrecAvg	18.2	1.2	1.2	0.7	0.7	0.7	1.1	2.2	3.1	2.1	2.3	1.6	1.1	
	PrecMax	24.7	3.0	3.0	2.2	2.1	1.7	2.0	3.9	5.9	3.5	3.9	4.6	2.3	
	PrecMin	12.4	0.1	0.1	0.1	0.1	0.0	0.5	0.7	0.7	0.4	0.7	0.3	0.1	
	PrecStd	3.1	0.7	0.8	0.6	0.6	0.4	0.4	0.7	1.3	0.7	0.8	1.0	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	32.3	34.0	36.9	45.7	60.7	64.2	61.4	55.0	48.4	37.1	32.1	
		MCWB	32.7	31.4	31.7	33.8	37.2	51.8	55.7	56.3	49.8	42.4	36.0	31.1	
	2%	DB	28.7	29.6	30.2	34.4	41.3	54.4	60.9	56.8	52.7	43.1	35.5	29.9	
		MCWB	27.8	29.4	29.3	31.7	35.4	47.8	54.9	52.9	49.2	40.2	34.3	29.1	
	5%	DB	24.9	26.6	27.0	32.4	38.3	50.8	56.6	54.5	51.1	40.4	32.7	27.2	
		MCWB	23.4	25.7	25.6	29.4	33.2	46.3	52.4	51.4	47.9	38.2	31.3	26.2	
	10%	DB	19.0	20.3	22.0	28.4	36.6	46.5	54.1	52.4	48.4	38.0	29.7	21.4	
		MCWB	18.0	19.6	20.7	26.6	32.2	42.7	50.6	49.6	46.0	36.0	28.6	20.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	31.4	32.3	34.6	40.9	53.2	57.7	57.2	51.3	43.7	36.2	31.1
			MCDB	33.4	31.8	33.6	36.2	43.8	58.2	62.5	60.8	53.6	47.5	36.7	31.4
2%		WB	28.1	29.3	29.2	32.8	37.4	49.9	55.4	53.7	49.9	40.7	33.7	28.9	
		MCDB	28.7	29.5	30.0	34.3	39.4	53.8	59.7	56.2	52.2	43.0	35.1	29.4	
5%		WB	23.9	25.9	25.7	30.2	35.6	46.4	53.0	51.7	48.0	38.5	31.3	26.3	
		MCDB	25.4	26.5	26.9	31.5	37.2	49.4	56.2	54.3	50.1	40.4	32.6	27.5	
10%		WB	18.6	19.6	21.1	26.6	34.2	43.7	50.5	49.9	46.3	36.5	28.3	21.2	
		MCDB	19.5	20.3	22.3	28.0	35.7	46.6	53.7	52.2	48.1	38.2	29.2	21.9	
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.6	11.0	11.4	7.9	7.0	7.0	5.8	5.9	7.0	9.1	9.1	
		MCDBR	13.6	13.6	10.5	12.5	10.1	12.6	11.9	8.4	7.0	6.5	7.6	10.5	
		MCWBR	12.5	13.3	10.0	11.1	7.6	8.5	7.9	6.1	4.9	6.0	7.1	10.6	
	5% WB	MCDBR	12.7	13.6	10.0	12.0	8.7	11.2	10.9	8.0	6.1	6.0	7.6	9.8	
		MCWBR	12.3	13.5	9.8	10.9	7.0	7.9	7.9	6.2	4.7	5.9	7.1	10.0	
Clear-Sky Solar Irradiance	taub	0.160	0.224	0.253	0.323	0.354	0.343	0.363	0.371	0.311	0.260	0.358	N/A		
	taud	1.899	2.049	2.171	1.989	2.029	2.398	2.450	2.374	2.505	2.232	2.821	N/A		
	Ebn at Noon	36	181	250	254	260	272	261	242	234	166	47	0		
	Edh at Noon	2	15	24	40	43	31	29	28	19	12	4	0		
All-Sky Solar Radiation	RadAvg	10	162	707	1409	1784	1885	1474	1004	575	176	23	0		
	RadStd	1	20	52	75	186	143	134	123	55	25	3	0		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page